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CFA: Self-Oriented Perfectionism

Data: Williams & Edwards (2021; OSF m5dy6)

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```
library(lavaan)
library(lavaangui)
library(psych)
library(dplyr)
library(semTools)
library(modelsummary)
library(tinytable)
```

Dataset: Self-Oriented Perfectionism Scale (SOP)

```
we_path <- here::here("data", "we_data.csv")
if (!file.exists(we_path)) {
    download.file("https://osf.io/download/6mdb4/", we_path)
}
we_raw <- read.csv(we_path)
we_raw <- dplyr::rename(we_raw, Att_check = SOP_6)
# Exclude: under-18, non-students, >6 missing study items, failed attention
check, duplicates
we_vars <- c(
    paste0("PASS_", c(1:9, 16:18)),
    paste0("NGSE_", 1:8),
```

```
    paste0("SOP_",  1:5)
)
we_raw <- we_raw[is.na(we_raw$Age) | we_raw$Age != 1, ]
we_raw <- we_raw[is.na(we_raw$Student) | we_raw$Student == 1, ]
we_raw$nmiss <- rowSums(is.na(we_raw[, we_vars]))
we_raw <- we_raw[we_raw$nmiss <= 6, ]
we_raw <- we_raw[!is.na(we_raw$Att_check) & we_raw$Att_check == 5, ]
we_raw <- we_raw[!duplicated(we_raw$PID_deID), ]
```

```
sop_data <- we_raw[, paste0("SOP_", 1:5)]
nrow(sop_data)
```

```
[1] 575
```

This example uses data from a conceptual replication of Seo (2008), examining self-efficacy as a mediator between self-oriented perfectionism and academic procrastination (Williams & Edwards, 2021; OSF: osf.io/m5dy6).

We focus here on the **SOP** subscale, which measures **self-oriented perfectionism**—the tendency to set high personal standards and be self-critical when those standards are not met:

Items	Content
SOP_1–SOP_5	Self-oriented perfectionism (5 items)

Items use a **5-point Likert scale**. The scale is theoretically **unidimensional**: all five items reflect a single latent construct. Data were collected online (Prolific Academic); N = 623 recruited.

> ℹ Data Cleaning
>
> Following the preregistered exclusion criteria:
>
> 1. Non-students or participants under 18 were excluded.
> 2. Participants missing more than 6 items across the three main scales (PASS, NGSE, SOP) were excluded.
> 3. Participants who failed the attention check (SOP_6; correct response = 5) were excluded.
> 4. Duplicate Prolific IDs were removed.
>
> The `load-data` chunk above applies all four criteria, reducing the sample to approximately N = 480.

Descriptive Statistics

```
modelsummary::datasummary_skim(sop_data)
```

	Unique	Missing Pct.	Mean	SD	Min	Median	Max	Histogram
SOP_1	5	0	3.5	1.1	1.0	4.0	5.0	▂▃▅▂
SOP_2	5	0	3.3	1.2	1.0	3.0	5.0	▂▃▅▂
SOP_3	5	0	3.0	1.1	1.0	3.0	5.0	▂▅▃▁
SOP_4	5	0	3.0	1.2	1.0	3.0	5.0	▂▅▃▂
SOP_5	5	0	2.6	1.2	1.0	2.0	5.0	▂▅▃▁

Inter-item Correlations

```
nv <- ncol(sop_data)
cor_sop <- cor(sop_data, use = "pairwise.complete.obs")
bg <- rgb(as.integer(cor_sop > 0), 0, as.integer(cor_sop < 0),
          alpha = abs(cor_sop))
bg[upper.tri(cor_sop, diag = TRUE)] <- NA
modelsummary::datasummary_correlation(sop_data) |>
  tinytable::style_tt(i = 1:nv, j = 1:nv + 1, background = bg)
```

	SOP_1	SOP_2	SOP_3	SOP_4	SOP_5
SOP_1	1	.	.	.	.
SOP_2	.77	1	.	.	.
SOP_3	.67	.68	1	.	.
SOP_4	.68	.75	.74	1	.
SOP_5	.46	.55	.57	.61	1

> **ℹ Note**
>
> Blue cells = positive correlations, red = negative; intensity = magnitude. All five SOP items measure the same construct, so uniformly positive and moderately strong correlations are expected.

Dimensionality

```
psych::fa.parallel(sop_data, n.iter = 100,
                   main = "Parallel Analysis — SOP Items")
```

```
Parallel analysis suggests that the number of factors =  2  and the number
of components =  1
```

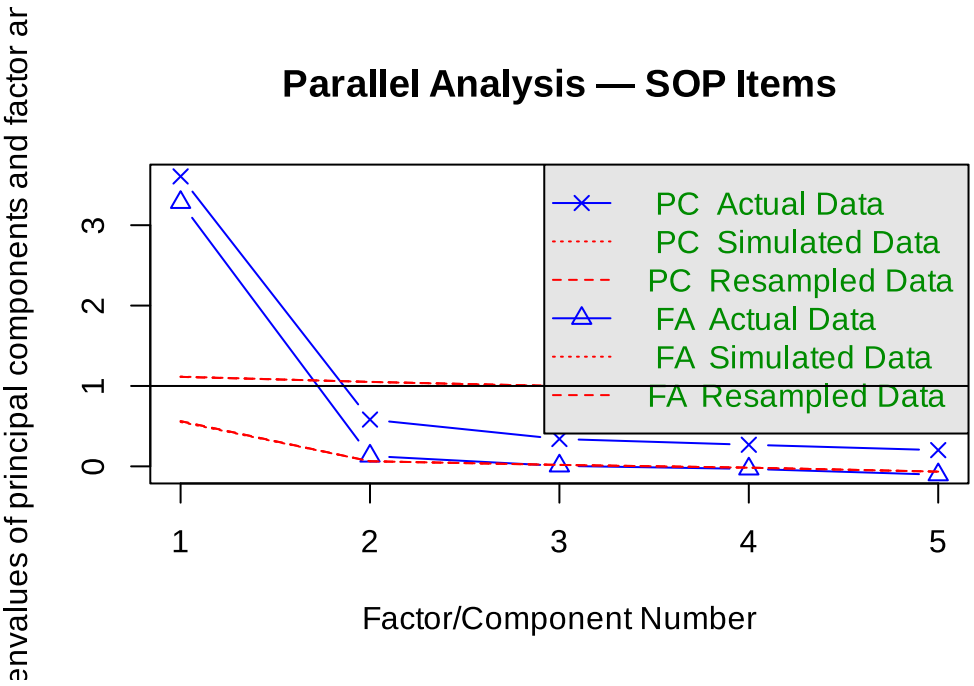

Figure 1: Scree plot with parallel analysis reference line.

```
ev <- eigen(cor_sop, only.values = TRUE)$values
data.frame(
  Factor        = seq_along(ev),
  Eigenvalue    = round(ev, 3),
  Cumulative_var = round(cumsum(ev) / sum(ev) * 100, 1)
)
```

```
  Factor Eigenvalue Cumulative_var
1      1      3.607           72.1
2      2      0.582           83.8
3      3      0.340           90.6
4      4      0.269           96.0
5      5      0.202          100.0
```

> **ℹ Note**
>
> Parallel analysis for a unidimensional scale should retain exactly one factor. A dominant first eigenvalue well above the parallel analysis reference confirms that the five items share a single common cause.

Confirmatory Factor Analysis

Example 1: Single-Factor CFA

[TODO: Identification strategies]

Because the SOP items are designed to reflect a single construct, we fit a **unidimensional (one-factor) CFA**. The latent factor variance is fixed to 1 (`std.lv = TRUE`) for identification, so all factor loadings are freely estimated.

```
model_1f <- "
  SOP =~ SOP_1 + SOP_2 + SOP_3 + SOP_4 + SOP_5
"
fit_1f <- cfa(model_1f, data = sop_data,
              std.lv    = TRUE,
              estimator = "MLR")
summary(fit_1f, fit.measures = TRUE, standardized = TRUE)
```

```
lavaan 0.6-21 ended normally after 17 iterations

  Estimator                                         ML
  Optimization method                           NLMINB
  Number of model parameters                        10

  Number of observations                           575

Model Test User Model:
                                              Standard      Scaled
  Test Statistic                                68.758      45.706
  Degrees of freedom                                 5           5
  P-value (Chi-square)                           0.000       0.000
  Scaling correction factor                                  1.504
    Yuan-Bentler correction (Mplus variant)

Model Test Baseline Model:

  Test statistic                              1868.666    1238.181
  Degrees of freedom                                10          10
  P-value                                        0.000       0.000
  Scaling correction factor                                  1.509

User Model versus Baseline Model:

  Comparative Fit Index (CFI)                    0.966       0.967
  Tucker-Lewis Index (TLI)                       0.931       0.934

  Robust Comparative Fit Index (CFI)                         0.967
  Robust Tucker-Lewis Index (TLI)                            0.934

Loglikelihood and Information Criteria:

  Loglikelihood user model (H0)              -3639.962   -3639.962
  Scaling correction factor                                  1.034
      for the MLR correction
```

```
  Loglikelihood unrestricted model (H1)      -3605.583   -3605.583
  Scaling correction factor                                  1.191
      for the MLR correction

  Akaike (AIC)                                7299.923    7299.923
  Bayesian (BIC)                              7343.467    7343.467
  Sample-size adjusted Bayesian (SABIC)       7311.721    7311.721

Root Mean Square Error of Approximation:

  RMSEA                                          0.149       0.119
  90 Percent confidence interval - lower         0.119       0.094
  90 Percent confidence interval - upper         0.181       0.146
  P-value H_0: RMSEA <= 0.050                    0.000       0.000
  P-value H_0: RMSEA >= 0.080                    1.000       0.994

  Robust RMSEA                                               0.146
  90 Percent confidence interval - lower                     0.109
  90 Percent confidence interval - upper                     0.186
  P-value H_0: Robust RMSEA <= 0.050                         0.000
  P-value H_0: Robust RMSEA >= 0.080                         0.998

Standardized Root Mean Square Residual:

  SRMR                                           0.032       0.032

Parameter Estimates:

  Standard errors                             Sandwich
  Information bread                           Observed
  Observed information based on                Hessian

Latent Variables:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
  SOP =~
    SOP_1             0.897    0.034   26.031    0.000    0.897    0.816
    SOP_2             1.087    0.033   32.881    0.000    1.087    0.874
    SOP_3             0.921    0.034   26.776    0.000    0.921    0.819
    SOP_4             1.052    0.032   32.503    0.000    1.052    0.871
    SOP_5             0.782    0.042   18.532    0.000    0.782    0.653

Variances:
                   Estimate  Std.Err  z-value  P(>|z|)   Std.lv  Std.all
   .SOP_1             0.405    0.032   12.713    0.000    0.405    0.335
   .SOP_2             0.366    0.040    9.052    0.000    0.366    0.236
   .SOP_3             0.415    0.038   10.900    0.000    0.415    0.328
   .SOP_4             0.352    0.039    9.049    0.000    0.352    0.241
```

```
   .SOP_5             0.824    0.052   15.898    0.000    0.824    0.574
    SOP               1.000                               1.000    1.000
```

```
lavaangui::plot_lavaan(fit_1f)
```

Model Fit

Modification Indices

Modification indices (MI) flag parameters whose addition would most improve model fit. For a unidimensional model, the most commonly suggested modifications are **residual covariances** between pairs of items: these indicate item-specific shared variance beyond the common factor (e.g., similar wording, shared method).

```
modindices(fit_1f, sort. = TRUE, minimum.value = 5)
```

```
     lhs op   rhs     mi    epc sepc.lv sepc.all sepc.nox
12 SOP_1 ~~ SOP_2 55.350  0.190   0.190    0.494    0.494
15 SOP_1 ~~ SOP_5 23.431 -0.137  -0.137   -0.238   -0.238
16 SOP_2 ~~ SOP_3 22.901 -0.125  -0.125   -0.321   -0.321
14 SOP_1 ~~ SOP_4 17.579 -0.104  -0.104   -0.276   -0.276
21 SOP_4 ~~ SOP_5 11.492  0.099   0.099    0.183    0.183
19 SOP_3 ~~ SOP_4 10.477  0.082   0.082    0.215    0.215
20 SOP_3 ~~ SOP_5  6.605  0.074   0.074    0.127    0.127
```

> **Note**
>
> An MI ≥ 10 (with $|\ epc\ | \geq 0.10$ and theoretical justification) is a common threshold for considering a modification. Adding residual covariances post hoc is exploratory and should be cross-validated or justified on substantive grounds.

Correlation Residuals

```
res_cor <- lavResiduals(fit_1f, type = "cor")$cov
nv_res  <- nrow(res_cor)
bg_res  <- rgb(as.integer(res_cor > 0), 0, as.integer(res_cor < 0),
               alpha = abs(res_cor))
bg_res[upper.tri(res_cor, diag = TRUE) | abs(res_cor) <= 0.1] <- NA
res_df <- as.data.frame(round(res_cor, 3))
tinytable::tt(cbind(" " = rownames(res_df), res_df)) |>
  tinytable::style_tt(i = 1:nv_res, j = 1:nv_res + 1, background = bg_res)
```

	SOP_1	SOP_2	SOP_3	SOP_4	SOP_5
SOP_1	0.000	0.055	−0.001	−0.032	−0.075

	SOP_1	SOP_2	SOP_3	SOP_4	SOP_5
SOP_2	0.055	0.000	−0.035	−0.008	−0.017
SOP_3	−0.001	−0.035	0.000	0.024	0.039
SOP_4	−0.032	−0.008	0.024	0.000	0.041
SOP_5	−0.075	−0.017	0.039	0.041	0.000

> **Note**
>
> Colored cells show residual correlations > |0.10|. Large residuals identify pairs of items whose observed correlation is not well reproduced by the single-factor model.

Goodness of Fit

```
lavaan::fitmeasures(fit_1f,
  c("cfi.robust", "tli.robust", "rmsea.robust",
    "rmsea.ci.lower.robust", "rmsea.ci.upper.robust",
    "srmr"))
```

```
           cfi.robust            tli.robust          rmsea.robust
                0.967                 0.934                 0.146
rmsea.ci.lower.robust rmsea.ci.upper.robust                  srmr
                0.109                 0.186                 0.032
```

Model Summary Table

```
glance_custom.lavaan <- function(x, ...) {
  m <- lavaan::fitmeasures(x)
  tibble::tibble(
    cfi.robust   = m[["cfi.robust"]],
    tli.robust   = m[["tli.robust"]],
    rmsea.robust = m[["rmsea.robust"]]
  )
}

gm <- data.frame(
  raw   = c("nobs", "statistic", "chi2.df", "p.chi2",
            "cfi.robust", "tli.robust", "rmsea.robust", "srmr"),
  clean = c("N", "Chisq", "df", "p(Chisq)",
            "CFI*", "TLI*", "RMSEA*", "SRMR"),
  fmt   = c(0, 2, 0, 3, 3, 3, 3, 3)
)
modelsummary::msummary(
    list("1-Factor" = fit_1f),
    fmt = fmt_decimal(digits = 2),
```

```
    estimate = "{estimate} [{conf.low}, {conf.high}]",
    statistic = NULL,
    metrics = "all",
    gof_map = gm
)
```

	1-Factor
SOP = SOP_1	0.90 [0.83, 0.96]
SOP = SOP_2	1.09 [1.02, 1.15]
SOP = SOP_3	0.92 [0.85, 0.99]
SOP = SOP_4	1.05 [0.99, 1.12]
SOP = SOP_5	0.78 [0.70, 0.86]
N	575
Chisq	68.76
df	5
p(Chisq)	0.000
CFI*	0.967
TLI*	0.934
RMSEA*	0.146
SRMR	0.032

ℹ Note

* CFI, TLI, and RMSEA use the Yuan–Bentler T_2^* robust correction (`.robust`), preferred over the Satorra–Bentler `.scaled` variants (Savalei & Rosseel, 2023).

Composite Reliability

Composite reliability (McDonald's ω) for the SOP scale:

```
semTools::compRelSEM(fit_1f)
```

```
$SOP

Composite `SOP` is composed of observed variables:
    SOP_1, SOP_2, SOP_3, SOP_4, SOP_5
True-score variance is represented by common factor(s):
    SOP
Total variance of composite `SOP` determined from the unrestricted model.
The proportion attributable to "true" scores is its model-based estimate of
```

```
reliability ("omega"):

[1] 0.905
```

Reporting

Sample Write-up

Method (Measures and Analysis)

> Self-oriented perfectionism was assessed with five items from the SOP scale (Williams & Edwards, 2021), rated on a 5-point Likert scale. We fit a unidimensional confirmatory factor analysis (CFA) model using the `lavaan` package (v0.6.21; Rosseel, 2012) in R. The latent factor variance was fixed to 1 for identification (`std.lv = TRUE`). To account for mild non-normality in the item responses, we used robust maximum likelihood estimation (MLR) with Yuan–Bentler scaled standard errors and a Satorra–Bentler scaled χ^2 test statistic. Missing data were handled with full information maximum likelihood (FIML). Model fit was evaluated using the robust CFI ($\geq .95$), robust RMSEA ($\leq .06$), and SRMR ($\leq .08$; Hu & Bentler, 1999). Composite reliability was estimated using McDonald's ω via `semTools::compRelSEM()`.

Results

> The hypothesized unidimensional CFA model showed acceptable fit to the data, robust $\chi^2(5) = 45.71$, $p < .001$, robust CFI = 0.967, robust RMSEA = 0.146, 90% CI [0.109, 0.186], SRMR = 0.032. Inspection of modification indices and correlation residuals did not reveal any theoretically justified model modifications. All five items loaded significantly on the SOP factor, with standardized loadings ranging from 0.65 to 0.87. Composite reliability was ω = 0.905, indicating good internal consistency.

References

Williams, M. N., & Edwards, S. R. (2021). Conceptual replication of Seo (2008): Self-efficacy as a mediator in the relationship between self-oriented perfectionism and academic procrastination. Current Issues in Personality Psychology, 10(2), 98–111. https://doi.org/10.5114/cipp.2021.110948

Kline, R. B. (2023). Principles and practice of structural equation modeling (5th ed.). Guilford Press.

Savalei, V., & Rosseel, Y. (2023). Computational options for standard errors and test statistics with incomplete normal and nonnormal data in SEM. Structural Equation Modeling, 30(2), 171–185. https://doi.org/10.1080/10705511.2022.2141754

Seo, E. H. (2008). Self-efficacy as a mediator in the relationship between self-oriented perfectionism and academic procrastination. Social Behavior and Personality, 36(6), 753–764. https://doi.org/10.2224/sbp.2008.36.6.753